

A Simplified Image Analysis Pipeline
for Detecting
sidekick Mutant Phenotypes
in *Drosophila* Gastrulation

Ethan Wang (jw2447@cam.ac.uk)

The Sanson Laboratory
Department of Physiology, Development and Neuroscience
University of Cambridge

August 2026

Declaration of Originality

I confirm that the material in this Research Project Report is not copied from any published material, nor is it a paraphrase or abstract of any published material unless it is identified as such and a full source reference is given.

I confirm that, other than where indicated as above, this document is my own work.

Author: Ethan Wang

CRSid: jw2447

Signature: Ethan Wang

Date: August 2026

1 Abstract

Sidekick (Sdk) is an evolutionarily conserved transmembrane cell-adhesion molecule of the immunoglobulin (Ig) superfamily. While vertebrate homologues (Sdk-1 and Sdk-2) regulate retinal synaptic connectivity and maintain kidney podocyte integrity, *Drosophila* Sdk is best known for driving ommatidial differentiation and visual circuit assembly. However, its role in broader epithelial morphogenesis remains incompletely understood.

While *sdk* null mutants are viable, the loss of both maternal and zygotic Sdk leads to distinct mechanical and morphogenetic defects in epithelial cell packing within a specific developmental window during germband extension. However, conventional phenotypic quantification remains exceptionally labour-intensive, requiring continuous, multi-metric measurements throughout the time frame in which the phenotype appears.

To streamline and accelerate this classification procedure, I acquired live-imaging movies and fixed tissue images from *Drosophila* embryos across multiple genotypes, and developed an automated image analysis pipeline combining ImageJ macro scripting, established machine learning tools, and custom Python scripts. Unexpectedly, applying this pipeline revealed that a standard wild-type strain expressing DE-cad::GFP displays a pronounced mutant-like phenotype, suggesting that fluorophore tagging of cell-adhesion molecules may subtly compromise endogenous Sdk function or junctional mechanics. Furthermore, this quantitative analysis established that fixed tissue preparations are unsuitable for evaluating these mutant phenotypes due to the loss of temporal context, highlighting the necessity of dynamic live-tissue analysis.

2 Introduction

Sidekick (Sdk) is a transmembrane cell-adhesion molecule belonging to the immunoglobulin (Ig) superfamily. All Sdk proteins possess a massive extracellular domain (> 2,000 amino acids) containing multiple Ig and fibronectin type III (FNIII) domains, alongside a short cytoplasmic tail ending in a conserved hexapeptide motif predicted to bind PDZ-domain scaffolding proteins (Fig. 1a). Its extracellular domain mediates homophilic adhesion through both *cis* (same-cell) and *trans* (opposite-cell) interactions.

The cell-adhesion molecule Sidekick (Sdk) belongs to the immunoglobulin (Ig) superfamily and plays critical roles in cell recognition and structural organization across species. In vertebrates, the homologues Sdk-1 and Sdk-2 are essential for establishing lamina-specific synaptic connections during retinal development and maintaining the functional filtration barrier in kidney podocytes. In *Drosophila*, early functional studies established Sdk as a key regulator of photoreceptor sorting during ommatidial development and a prerequisite for precise visual motion detection circuits.

Sdk is recognized as the first resident protein specifically localized to tricellular adherens junctions (Fig. 1b,c). Sdk localizes at tricellular apical vertices across various *Drosophila* epithelial tissues, including the embryonic ectoderm, amnioserosa, larval wing disc, and early follicular epithelium. Mosaic clone analyses show that Sdk expression from at least two of the three contacting cells is sufficient for homophilic recruitment of Sdk to a tricellular vertex. Furthermore, super-resolution imaging demonstrates that multicellular rosette centers (where five or more cells converge) are composed of distinct, spatially separable tricellular vertices marked by individual Sdk strings, rather than a single merged multicellular vertex.

While *sdk* null mutants are viable, the complete absence of maternal and zygotic Sdk leads to distinct mechanical and morphogenetic defects during germband extension (GBE). Specifically,

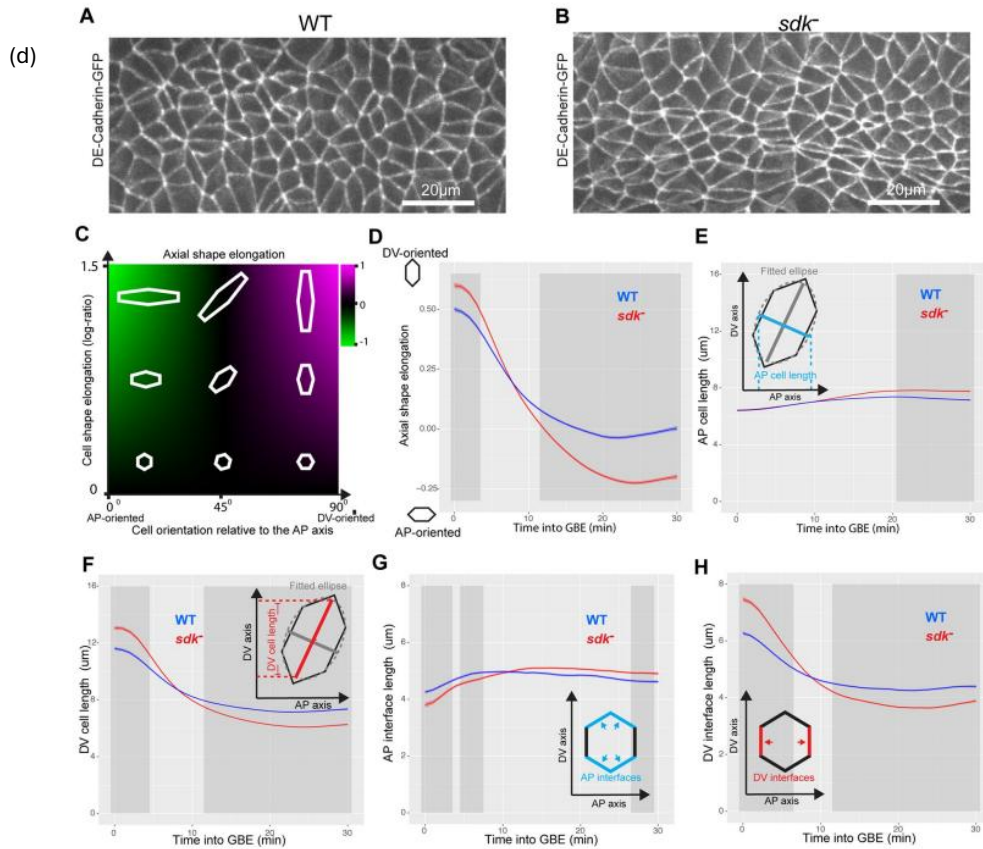
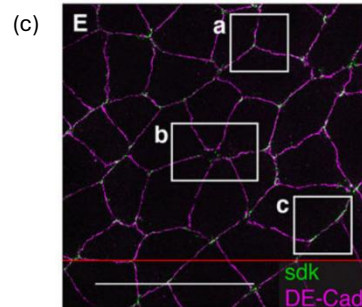
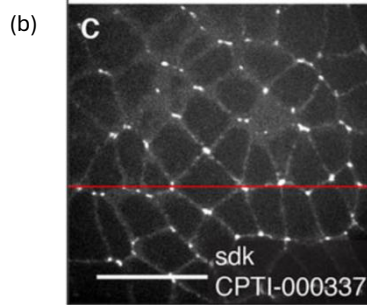
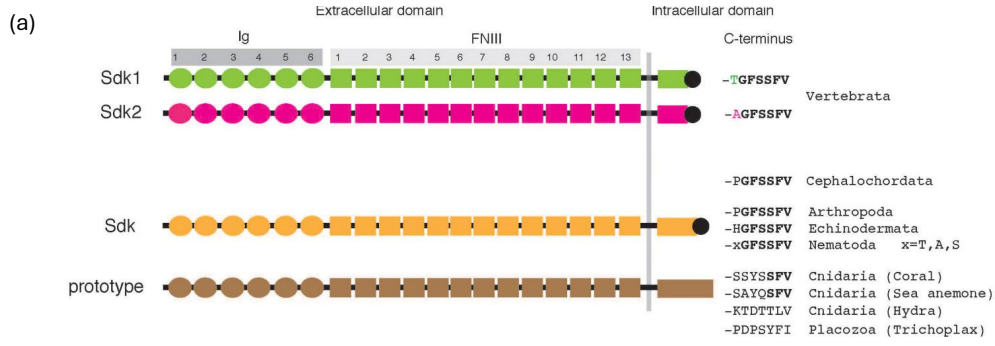


Figure 1: **Basic Information about Sidekick** (a) Predicted structure of vertebrate Sdk1 and Sdk2, and fly and worm Sdk proteins (Yamagata, 2020). (b), (c) Sdk localisation at tricellular junctions (Lye, Naylor and Sanson, 2014). (d) A,B: wildtype and *sdk* mutant cell packing phenotypes. C,D,E,F,G,H: mechanical measurements for wt and *sdk* over the period of GBE (Finegan et al., 2019).

ectodermal cells in *sdk* mutants display irregular, overly anisotropic apical shapes that become abnormally elongated along the anterior-posterior (AP) axis. Consequently, polarized cell intercalation contributing to tissue extension is diminished; T1 transitions (neighbour exchanges) are less efficiently oriented toward tissue extension, and their subsequent resolution phase is significantly delayed.

Previous classifications of wild-type (WT) and *sdk* mutants were primarily based on tracking the temporal dynamics of developing embryos (Fig. 1d). Although quantifying these dynamic trajectories provides a comprehensive demonstration of how mechanical and morphogenetic features evolve over time, the underlying cell-tracking and measurement processes are exceptionally labor-intensive and computationally demanding. To overcome these throughput limitations, developing a classification model trained exclusively on static, single-frame spatial data offers a rapid, highly scalable alternative. This approach can provide novel insights into the collective tissue architecture of both WT and *sdk* mutant embryos, directly complementing our understanding of Sdk-mediated morphogenetic mechanisms.

3 Results

The wild types and *sdk* null mutants in this paper refer to *w; sqh > GAP43 - mCherry/FM7* and *GAP43 - mCh, sdk⁵⁰⁵⁴/CyO* respectively, unless otherwise stated.

3.1 wt and *sdk* mutants have similar average cell shapes

To investigate whether individual epithelial cells in wild-type (WT) and *sdk* mutant *Drosophila* embryos display distinct intrinsic cell geometries, I applied **ShapeEmbedLite**, a machine-learning framework designed to extract low-dimensional latent shape embeddings from individual cell contours and distance matrices (Fig. 1a). By encoding cell outlines independently of spatial rotation or scale, this framework establishes an unbiased evaluation of cell morphology beyond conventional scalar metrics such as cell area or perimeter.

Unsupervised dimensionality reduction of the resulting latent feature space revealed no distinct separation between control and mutant cell populations. Multi-dimensional projections generated via Principal Component Analysis (PCA; Fig. 1d), Uniform Manifold Approximation and Projection (UMAP; Fig. 1e), and *t*-Distributed Stochastic Neighbor Embedding (*t*-SNE; Fig. 1f,g) consistently showed that control (blue) and *sdk* mutant (orange) data points intermingle homogeneously. The lack of discrete clustering across these latent spaces indicates that the overall distribution of individual cell shapes in *sdk* mutants is broadly indistinguishable from that of wild-type controls.

To test whether cell shape features could distinguish the two genotypes, supervised binary classification models were trained directly on the latent space representations (Fig. 1h). Overall, the classifiers performed poorly, scoring only slightly above the 0.50 baseline expected for random guessing on a balanced dataset. During cross-validation, model performance remained near chance levels, with the Logistic Regression model yielding a mean F_1 score of 0.541 ± 0.037 and the Random Forest model a mean F_1 score of 0.480 ± 0.033 . Results on the test dataset were similarly weak: Logistic Regression achieved a mean F_1 score of 0.618 ± 0.009 , while Random Forest reached a mean F_1 score of 0.611 ± 0.035 . Together, these results confirm that average latent shape features do not contain enough clear information to reliably tell wild-type and *sdk* mutant embryos apart (Fig. 1c); the loss of Sdk does not significantly alter mean single-cell morphology.

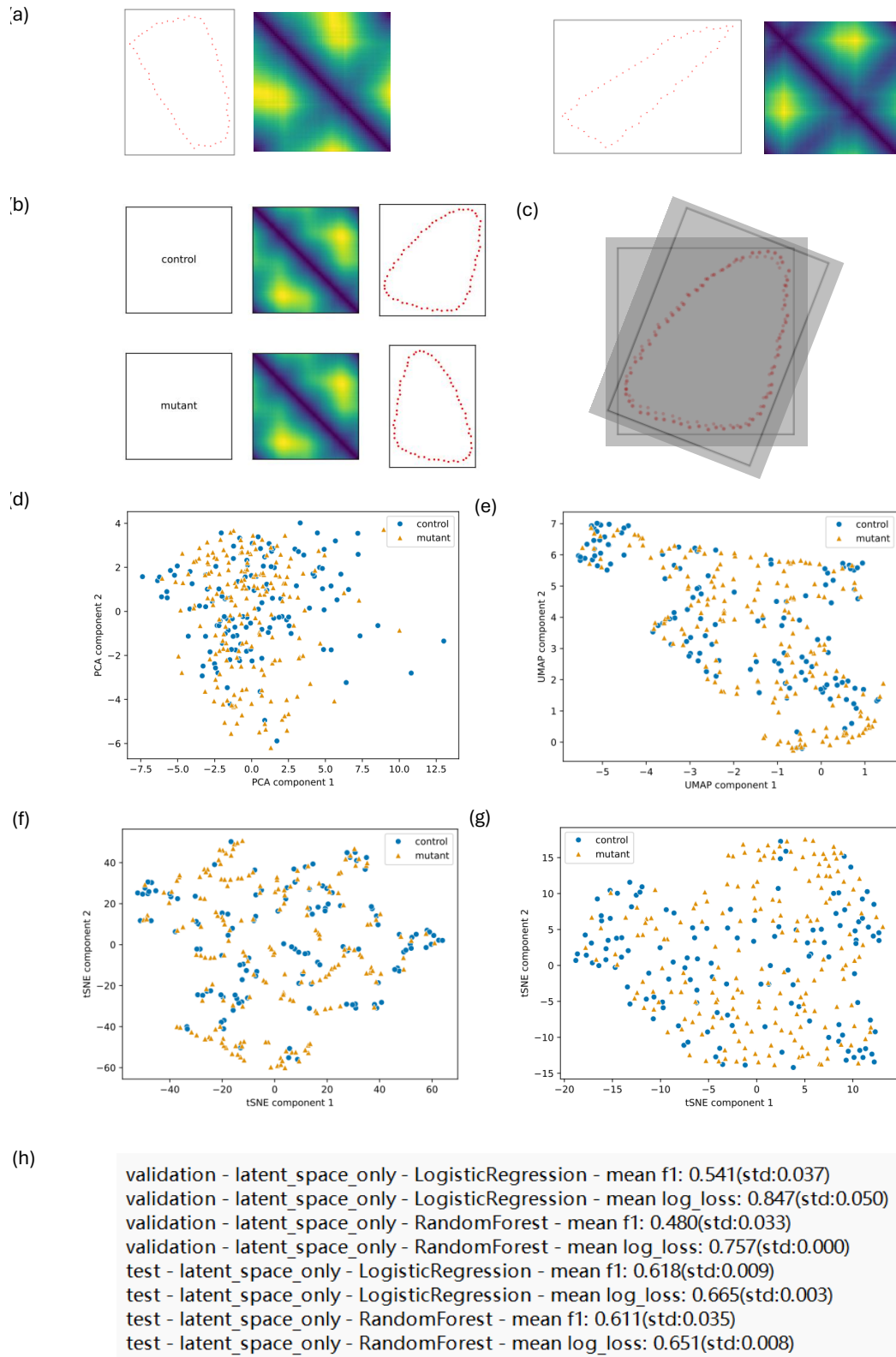


Figure 2: **ShapeEmbedLite Results** (a) Contours and distance matrices of two cells. (b) Average cell shapes of wt and *sdk* mutant. (c) Overlay of average cell shapes of wt and *sdk* embryos. (d), (e), (f), (g) PCA, UMAP, tSNE-05 and tSNE-p30 of wt and *sdk* embryos. (h) Validation and test scores.

3.2 Random Forest Classifies Wild-Type and *sdk* Mutants

Acquiring and processing high-resolution embryo datasets is inherently labor-intensive and time-consuming, resulting in a small sample size ($N = 13$ embryos). Under such sample constraints, complex deep learning architectures are prone to overfitting and poor generalization. Consequently, I resorted to classical machine learning approaches, beginning with a Random Forest (RF) classifier. Crucially, each input data point supplied to the model represents an individual single embryo rather than an isolated cell. By structuring the input at the single-embryo level, the model is able to ask tissue-scale questions and evaluate global structural patterns that single-cell analysis fails to detect.

Using a Leave-One-Out cross-validation (LOOCV) scheme to rigorously evaluate performance on small sample sizes, the RF model correctly classified all 13 embryo samples, achieving 100% prediction accuracy (Fig. 2a). This indicates that whole-embryo summary features contain consistent signatures that separate wild-type controls from *sdk* mutants.

Analysis of the feature importances extracted from the model indicates that cell orientation angles serve as the primary criteria for grouping the embryos (Fig. 2b). Specifically, multiple angle distribution bins (including `Angle_bin_2`, `Angle_bin_6`, and `Angle_bin_9`) together with cell area (`Area_bin_2`) accounted for the majority of the model’s predictive weight.

To mitigate the risk of overfitting inherent to high feature-to-sample ratios ($p = 64, N = 13$), I tested whether full high-dimensional input was necessary by retraining the classifier exclusively on these top-ranked feature bins. This feature reduction yielded identical 100% LOOCV accuracy with nearly identical decision boundaries, demonstrating that tissue-scale discrimination is driven by a compact, highly informative subset of morphological parameters rather than sparse high-dimensional noise.

Examining the underlying angle distribution histograms explains why these retained features are so informative (Fig. 2e). Wild-type embryos consistently exhibit a unimodal distribution centered around intermediate angles ($\sim 90^\circ$), indicating regular cell alignment relative to tissue axes. In contrast, *sdk* mutant embryos display a bimodal distribution with prominent peaks near 0° and 180° . This shift reflects the abnormal cell elongation along the anterior-posterior (AP) axis characteristic of the *sdk* phenotype.

3.3 Support Vector Machine Achieves Clean Separation in Feature Space

To complement the Random Forest results, I applied a Support Vector Machine (SVM) to test whether wild-type and *sdk* mutant embryos are linearly separable in feature space. Like the Random Forest classifier, every input vector fed into the SVM represents a single embryo, enabling the model to construct decision boundaries based on tissue-scale structural properties. SVMs are particularly well-suited for small datasets because they construct a maximum-margin decision boundary based on support vectors, minimizing overfitting when working with high-dimensional feature representations.

Similar to the Random Forest model, LOOCV evaluation of the full SVM classifier yielded 100% classification accuracy across all 13 samples (Fig. 2c). Feature weighting similarly highlighted key cell area and angle distribution bins as the primary contributors to class distinction (Fig. 2d). Restricting the SVM input to only these top-weighted feature bins likewise preserved 100% LOOCV classification accuracy, confirming the parsimony and robustness of the reduced feature representation across distinct classifier architectures.

To visualize how the model separates the two groups, the decision boundary was projected onto 2-dimensional spaces. Projecting the dataset onto the first two principal components (Fig. 2g) reveals a wide, clean margin of separation between wild-type and *sdk* mutant clusters without overlap. A similarly clear boundary is observed when plotting the decision line directly against the two features with the highest importances, namely *Angle_bin_6* and *Area_bin_2* (Fig. 2h). Together, these results confirm that whole-embryo topological features provide a robust, linearly separable signature that reliably distinguishes *sdk* mutants from wild-type controls.

3.4 DE-cad::GFP Wild-Type Displays an Intermediate Mutant-Like Phenotype

To evaluate how well the classification models generalize across different imaging markers, I tested both the Random Forest and Support Vector Machine models on an independent dataset expressing DE-cad::GFP, comprising both wild-type (DE-cad::GFP) and mutant (DE-cad::GFP; *sdk*⁵⁰⁵⁴) backgrounds. As expected, the DE-cad::GFP; *sdk*⁵⁰⁵⁴ embryos were consistently classified as mutants. However, both models unexpectedly classified a large proportion of the DE-cad::GFP wild-type embryos as being mutant-like. Quantitatively, the cell orientation angle distributions of these misclassified wild-type embryos closely mirrored those of true mutants, displaying a bimodal profile with a pronounced dip at intermediate angles and elevated frequencies near 0° and 180° (Fig 3f).

Visual inspection of the pre-processed tissue images explained the model’s prediction while revealing key structural nuances. Individual cells within the DE-cad::GFP wild-type tissue do indeed exhibit abnormal elongation along the anterior-posterior (AP) axis, providing the tissue-scale metric that drove the automated classification. Crucially, however, these elongated cells appear in isolation across the tissue rather than converging into higher-order structural hallmarks of *sdk* phenotypes, such as tissue-wide cell "ladders" or persistent, unresolved multicellular rosettes (Fig. 3i,j). Together, these findings suggest that fluorophore tagging of endogenous E-cadherin is not entirely neutral, but rather induces a mild intermediate phenotype positioned between an untagged wild-type baseline and an *sdk* null mutant.

3.5 Fixed Tissue Preparations Fail to Capture Mutant Phenotypes Across Genotypes

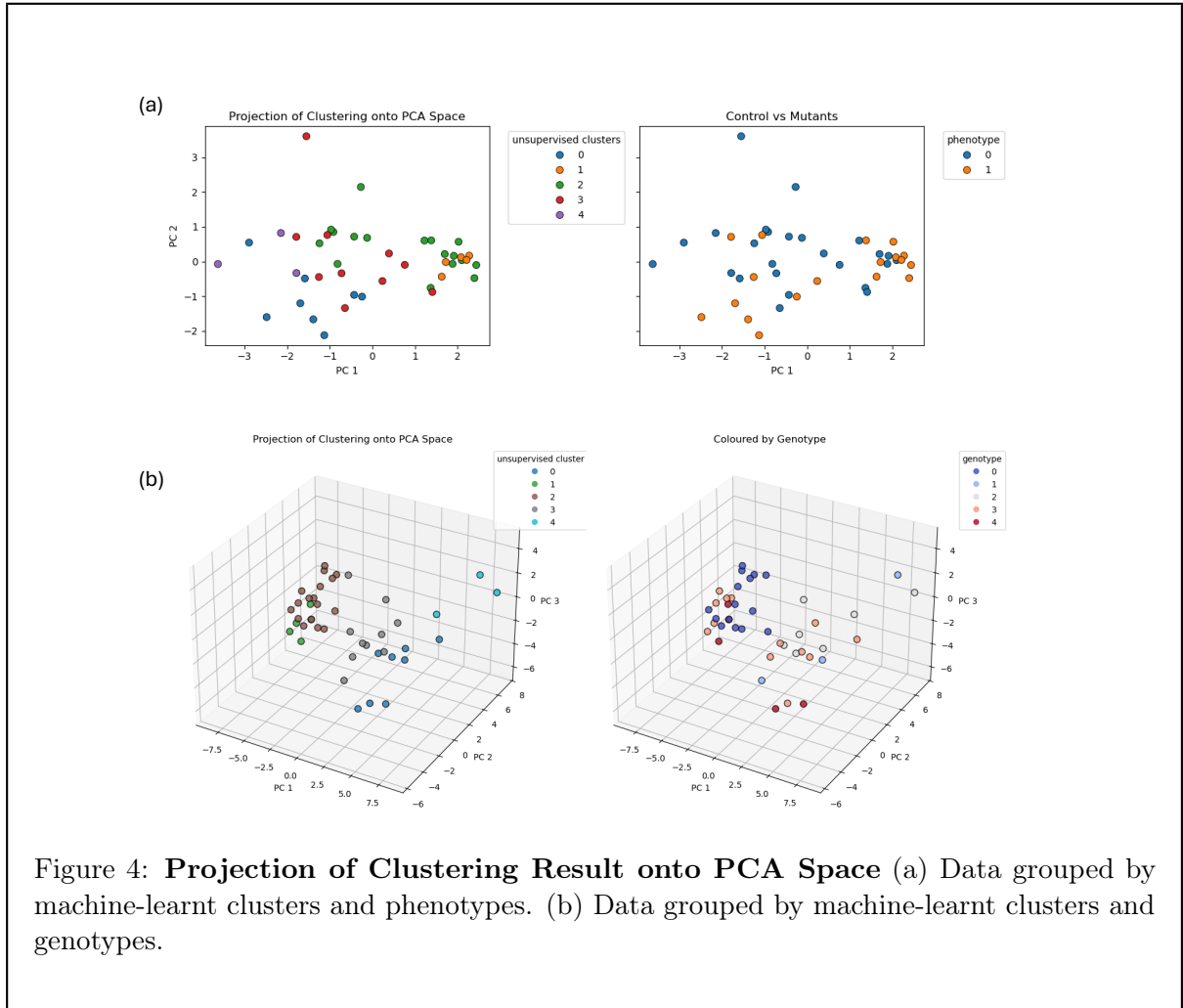
To test whether fixed tissue preparations can be used for automated phenotypic classification, I analyzed fixed embryos at stage 7 and stage 8 across four distinct genotypes: two wild-type controls (*yw* and *del7*) and two *sdk* null mutant alleles (*sdk*⁵⁰⁵⁴ and *del15*). Unexpectedly, when evaluated by both the Random Forest and Support Vector Machine models, nearly all fixed embryos were classified as wild-type, regardless of their actual genotype.

This widespread misclassification highlights a major limitation of using fixed tissue preparations to quantify Sdk-mediated defects. Because *sdk* mutants display a dynamic and transient phenotype during germband extension, static fixed images only capture isolated snapshots of tissue topology. Consequently, many fixed samples fall outside the narrow temporal window in which morphogenetic defects peak.

3.6 Unsupervised Clustering Features No Correspondence to Genotype

Given that the binary decision models (Random Forest and SVM) failed to separate fixed-tissue genotypes into wild-type and mutant categories, I tested whether the high-dimensional feature space contained intrinsic cluster structures that aligned with genetic background. Using k -means clustering on whole-embryo feature vectors, an elbow plot identified $k = 5$ as the optimal number of clusters, matching the total number of genotypes evaluated across live and fixed experiments.

However, comparing the machine-learned cluster assignments to true sample identities revealed no meaningful alignment with genotype or phenotype. This lack of correlation demonstrates that structural feature variance in fixed tissue is predominantly driven by sample-to-sample preparation noise rather than genotype-specific morphogenetic signatures.



3.7 Cell Neighbour Distributions Remain Invariant Across Genotypes and Stages

During germband extension, tissue elongation is driven by polarized cell intercalation via T1 transitions (neighbour exchanges) rather than cell division. Because these local rearrangements directly alter cell-cell connectivity, I examined the overall neighbour count distributions between genotypes and across the GBE time window. To quantify local connectivity, I developed an automated neighbour-counting algorithm. Although the algorithm slightly underestimates neigh-

bour counts for cells with severe outline distortion, it normally recovers neighbour counts on segmented tissue images with high accuracy.

Plotted across multiple developmental stages (Stage 7, Stage 8, and Stage 9), the neighbour count distributions showed no apparent differences between wild-type controls (*yw*, *del7*) and *sdk* null mutants (*sdk*⁵⁰⁵⁴, *del15*) (Fig. ?). Across all genotypes and stages, the distributions consistently follow a near-Gaussian profile centered sharply between 5 and 6 neighbours per cell, representing standard hexagonal packing. This invariance demonstrates that loss of Sdk does not disrupt the fundamental topological distribution of cell neighbour counts, further indicating that fixed tissue metrics are insufficient for capturing *sdk* phenotypic defects.

Plotted across multiple developmental stages (Stage 7, Stage 8, and Stage 9), the neighbour count distributions showed no apparent differences between wild-type controls (*yw*, *del7*) and *sdk* null mutants (*sdk*⁵⁰⁵⁴, *del15*) (Fig. ?). Across all genotypes and stages, the distributions consistently follow a near-Gaussian profile centered sharply between 5 and 6 neighbours per cell, reflecting standard hexagonal packing. There are two potential explanations for this topological invariance. First, the loss of Sdk may simply have no effect on overall cell neighbour distributions, meaning that Sdk-mediated junctional defects disrupt local cell mechanics without altering global tissue connectivity. Second, because the *sdk* mutant phenotype is highly dynamic during germband extension, static fixed snapshots may catch embryos outside the narrow temporal window in which the defects are visible. In either case, these findings further indicate that fixed tissue metrics are insufficient for capturing *sdk* defects.

4 Discussion

Conventional methods for evaluating *sdk* null mutant phenotypes rely on labor-intensive, continuous tracking of cell geometry and tissue mechanics across entire developmental movies. In this study, I demonstrated that whole-embryo spatial profiles extracted from a single static image are sufficient to accurately distinguish wild-type controls from *sdk* mutants. Using classical machine learning algorithms—Random Forest and Support Vector Machine—trained on tissue-scale features such as cell orientation angle distributions, both models achieved 100% classification accuracy on live-imaging datasets despite a small sample size ($N = 13$). By bypassing the requirement for continuous time-series cell tracking, this approach drastically reduces data acquisition and processing overhead, providing a rapid, scalable framework for morphogenetic phenotype classification.

A crucial yet unexpected finding was that wild-type embryos expressing DE-cad::GFP were consistently classified as mutant-like by both models. Quantitative analysis confirmed that these embryos exhibit bimodal cell orientation angle distributions and elongation of isolated cells along the anterior-posterior (AP) axis. Together these two clues suggest that DE-caed::GFP wild types assume an intermediate phenotype between untagged wild-type controls and *sdk* null mutants. This result suggests that fluorophore tagging of endogenous E-cadherin is not entirely neutral, but may subtly compromise junctional mechanics or sterically interfere with Sdk recruitment at tricellular vertices. Future experiments systematically comparing different fluorophore insertion sites and alternative junctional tags will be important to determine how Cadherin tagging interacts with Sdk-mediated tissue packing.

In contrast to live imaging, applying automated classification, *k*-means clustering, and cell neighbour analysis to fixed tissue preparations across four genotypes (*yw*, *del7*, *sdk*⁵⁰⁵⁴, and *del15*) yielded uniformly wild-type profiles. This failure to detect mutant phenotypes in fixed

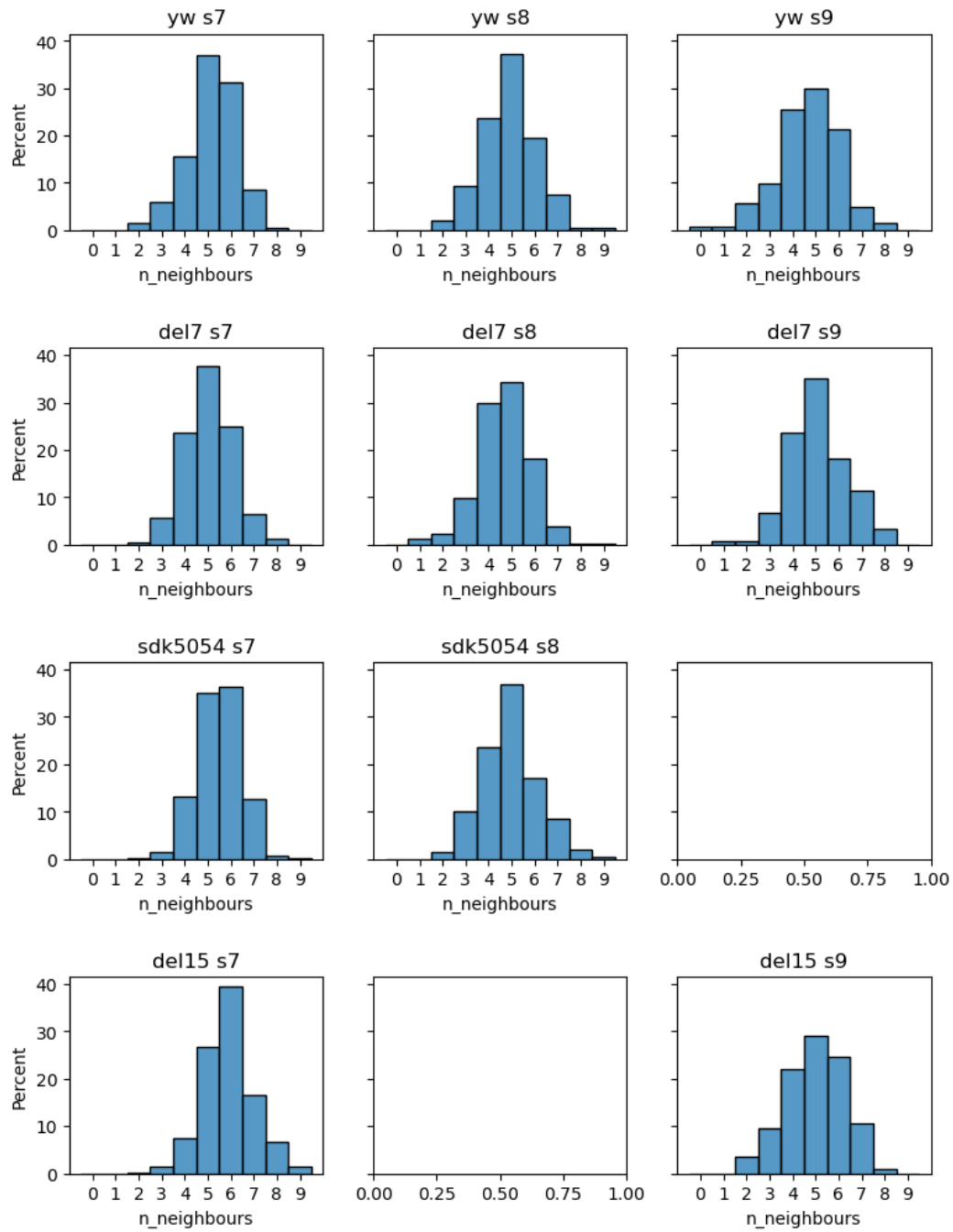


Figure 5: Neighbour number distribution for 4 genotypes across all stages

samples highlights critical constraints inherent to static tissue assays. Because the *sdk* phenotype is transient and tightly linked to active cell intercalation during germband extension, static fixed preparations frequently miss the narrow developmental window during which physical defects peak. Consequently, while fixed preparations offer high optical clarity, dynamic live-tissue imaging remains indispensable for quantifying Sdk-dependent morphogenetic processes.

5 Materials and Methods

5.1 Fly Strains

Genotype	Description	Origin	Stock ID	Phenotype
<i>yw</i> ⁶⁷	Standard wild-type control strain	Bloomington	1313	Wild-type
<i>sdk</i> ^{Δ7}	Control allele	Tressman	1444	Wild-type
<i>FRT19A, sdk</i> ^{Δ15}	Imprecise excision (1240 bp deletion at 5'-end; removes end of exon 1 and exon 2; see Astigarraga et al., 2018)	Tressman	1445	<i>sdk</i> null
<i>Mi{ET1}<i>sdk</i>^{MB05054}, <i>w</i>¹¹¹⁸</i>	Minos transposon insertion allele	Bloomington	1392	<i>sdk</i> null
<i>GAP43-mCh, sdk</i> ⁵⁰⁵⁴ / <i>CyO</i>	Recombinant from BS1401 (<i>w; sqh>GAP43-mCherry/FM7</i>) and BS1392	Sanson Lab	1572	<i>sdk</i> null (Red membrane)
<i>w; sqh>GAP43-mCherry/FM7</i>	Fluorescent membrane reporter line	S. De Renzis	1401	Wild-type (Red membrane)
<i>DE-cad::GFP</i>	<i>shg::GFP</i> knock-in reporter line (Huang et al., 2009); reselected by S. Ma	Huang et al.	1566	Wild-type (Green membrane)
<i>FRT19A, sdk</i> ^{Δ15} ; <i>DE-cad::GFP</i>	Recombinant line integrating fluorescent junctional reporter into null background	This study	—	<i>sdk</i> null (Green membrane)

5.2 Primers for PCR Genotyping

Primer Name	Sequence (5' → 3')	<i>T_m</i> (°C)
<i>sdk1a_r</i> (230)	CTCTCAAGACTCCATTGAGCTGTGGCTC	62.3
<i>sdk1b_r</i> (231)	CTCAGGACTCCATTAAGCTGCGGTTCTG	62.6
<i>sdk2a_r</i> (232)	GATGTAGCCTTGAAGCGGGCCATTC	62.0

5.3 Validation of the *sdk*^{del15} Deletion Allele

To confirm the genetic identity of the *sdk*^{del15} line maintained in our laboratory, genomic DNA was extracted and analyzed using a three-primer diagnostic PCR targeting the *sdk* locus. The reaction design utilized a single forward primer in combination with two reverse primers: one flanking the downstream end of the locus (R1) and a second positioned internally within the gene (R2). PCR products were resolved via agarose gel electrophoresis (Fig. 5a). Reactions using the internal R2 primer produced a light, truncated DNA of identical molecular weight in both *del15* and *DE-cad::GFP; del15* mutant samples, indicating the presence of the deletion. In contrast, the control strain (*DE-cad::GFP*) yielded a significantly heavier, full-length product corresponding to the intact wild-type *sdk* locus.

To verify sequence identity across the deletion junction, the amplified PCR products from the *del15* samples were purified via gel extraction and subjected to sequencing. The resulting sequence data confirmed that the laboratory *del15* strain harbors the exact deletion sequence of the established *sdh^{del15}* null allele.

5.4 IHC Staining

5.4.1 Risk assessments

The following risk assessments should be read and follow while undertaking this protocol:

- RA1 – Lab work
- HSD3 – Dechorionating embryos
- HSD5 – Staining with DAPI
- HSD6 – Formaldehyde fixation
- HSD8 – Immuno-staining
- HSD10 – Methanol devitalisation

5.4.2 Embryo Collection and Fixation

1. Collect embryos on apple juice agar plates from laying cages.
2. Wash embryos into a collecting mesh basket and dechorionate in thin domestic bleach (dilute sodium hypochlorite) for 1 min.
3. Rinse thoroughly with tap water to remove all traces of bleach.
4. Prepare a glass vial containing 3 mL of 37% formaldehyde (lower phase) and 3 mL of heptane (upper phase).
5. Gently dry the base of the embryo basket and use a fine brush to transfer embryos into the upper heptane layer. Embryos will sink to the heptane-formaldehyde interface.
6. Agitate the vial vigorously on a shaker for 5–10 min at room temperature (extend up to 20 min if lower formaldehyde concentrations are used).
7. Aspirate and discard the lower formaldehyde phase into dedicated hazardous liquid waste.

5.4.3 Methanol Devitellinization

1. Add 3 mL of 100% methanol to the heptane phase containing fixed embryos.
2. Shake the vial vigorously by hand for 30 s. Embryos that have successfully shed their vitelline membrane will sink to the bottom.
3. Carefully remove and discard the upper heptane layer into hazardous waste.
4. Add 1 mL of fresh methanol, shake briefly, and allow devitellinized embryos to settle.
5. Using a Pasteur pipette, collect the sunken embryos (discard any uncracked embryos remaining at the interface) and transfer them to a 1.5 mL microcentrifuge tube.
6. Wash twice with 100% methanol. *Note: Fixed embryos can be stored in methanol at -20°C for several months.*

5.4.4 Blocking and Primary Antibody Incubation

1. Wash embryos 3×15 min in PTX buffer on a nutator at room temperature.
2. Block non-specific binding sites by incubating in PTB buffer for 30 min at room temperature.
3. Remove PTB and add primary antibody diluted in 250 μ L of PTB (consult laboratory antibody index for target dilutions).
4. Incubate with gentle rocking overnight at 4°C.
5. **Optional Post-Fixation (for anti-Sqh or weak signal targets):** Aspirate primary antibody, wash once, and incubate in 4% formaldehyde in PBS for 20 min. Wash twice rapidly with PTX, followed by 2×15 min washes in PTX.

5.4.5 Secondary Antibody Incubation and Mounting

1. Add fluorophore-conjugated secondary antibody (typically 1:500 dilution in 500 μ L PTB).
2. Incubate on a rocker for 1 h at room temperature protected from light.
3. Rinse twice with PTX, followed by 2×15 min washes in PTX in the dark.
4. Equilibrate embryos in Vectashield mounting medium prior to selecting and orienting embryos on slides for confocal imaging.

5.4.6 References

Method adapted from Tara’s Thesis and used in Finegan et al, 2019.

5.4.7 Modifications

- Where a biotinylated secondary is used, repeat steps 2.6 and 2.7 with the SA-dye secondary.
- Tween-20 can be used in place of Triton X100 in the PTX at a 0.1% concentration. This gives a less stringent wash environment which improves signal from antibodies including Comm.

5.5 Live Image Processing and Feature Extraction

Time-lapse imaging datasets were spatially and temporally standardized prior to downstream analysis. Each movie was rotated to align the ventral midline horizontally near the top of the frame and reflected so that the anterior pole was oriented to the left and the posterior pole to the right. To ensure temporal consistency across samples while capturing the fully expressed mutant phenotype, a single frame corresponding to the first instance of cell division during germband extension was selected for processing. For Z-stack compilation, the surface focal plane of the embryo was identified, and a 5-slice sub-stack (comprising the surface plane, 1 slice above, and 3 slices below) was converted into a single 2D image using Maximum Intensity Projection.

To prevent geometric distortion caused by tissue involution along the midline, a rectangular region of interest (ROI) was cropped starting 1 to 2 cell layers below the midline, where cells retain representative lateral morphologies. Cell boundary segmentation within these cropped ROIs was performed using Cellpose, and the resulting cell outlines were exported as ROI files.

Individual segmented cells were extracted as binary mask images. For each cell, a suite of 2D morphological measurements—including Area, X, Y, Perim., Major, Minor, Angle, Circ., Feret,

FeretX, FeretY, FeretAngle, MinFeret, AR, Round and Solidity)—was extracted and compiled into CSV format. Additionally, binary cell masks were converted into distance matrices to serve as input for single-cell shape analysis (**ShapeEmbedLite**).

Explanations for Measurements Taken:

Area: area of selection

X: x coordinate of the centroid

Y: y coordinate of the centroid

Perim.: length of the outside boundary of the selection

Major: primary axis of the best fitting ellipse

Minor: secondary axis of the best fitting ellipse

Angle: the angle between the primary axis and a line parallel to the X-axis of the image

Circ.: $4\pi \times \frac{Area}{Perim.^2}$. A value of 1.0 indicates a perfect circle. As the value approaches 0.0, it indicates an increasingly elongated shape.

Feret: The longest distance between any two points along the selection boundary

FeretX: the x coordinate of the starting point of the Feret diameter

FeretY: the Y coordinate of the starting point of the Feret diameter

FeretAngle: the angle between the Feret diameter and a horizontal line

MinFeret: The shortest distance between any two points along the selection boundary

AR: $\frac{Major}{Minor}$, aspect ratio of the cell's fitted ellipse.

Round: $4 \times \frac{Area}{\pi \times Major^2}$, an indicator of the roundness of the cell.

Solidity: $\frac{Area}{ConvexArea}$

5.6 ShapeEmbedLite

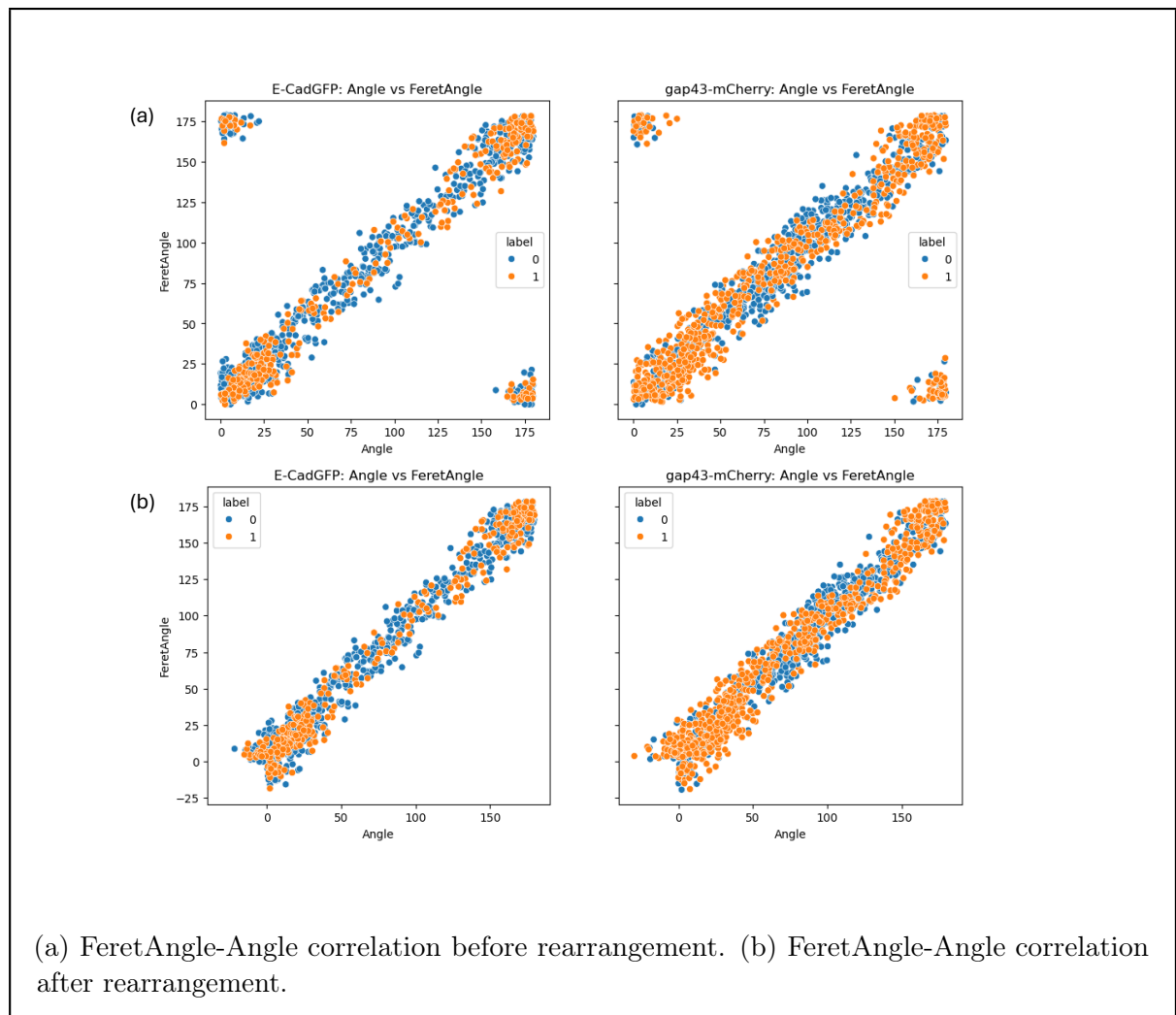
ShapeEmbed is a method for turning 2D outlines into compact computer codes (latent vectors) so an AI model can analyze or generate shapes regardless of how they are rotated, moved, or resized.

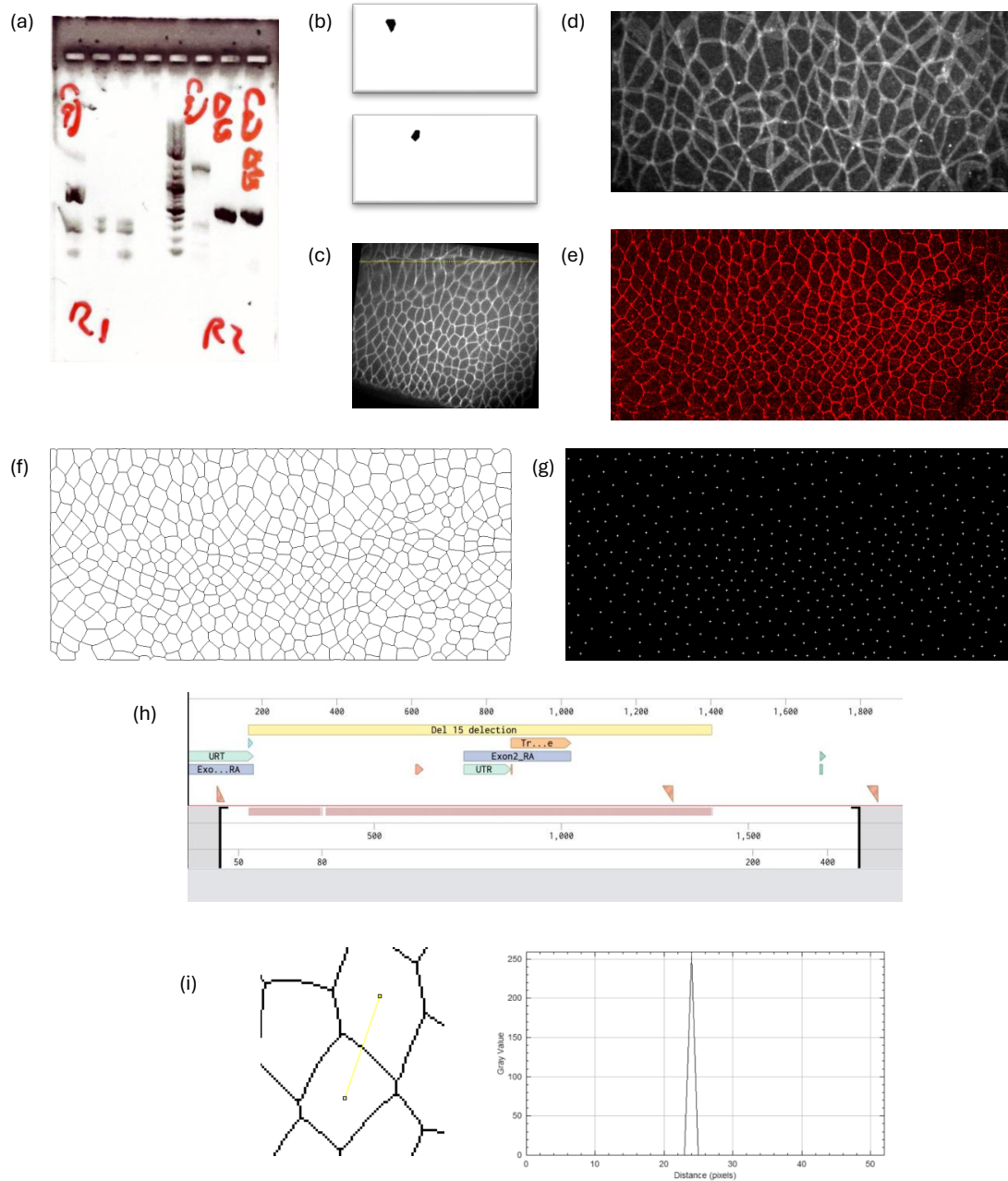
1. Instead of saving x, y coordinates (which change if you move or tilt an object), ShapeEmbed places 64 evenly spaced points around the shape's outline. It then creates a table measuring the distance from every point to every other point. Finally, it scales this distance table so the overall size of the shape does not matter.
2. A shape's outline could be traced starting at any point, or traced clockwise versus counterclockwise. ShapeEmbed uses a special AI model (a Variational Autoencoder with circular padding and reflection passes) that reads the distance table the exact same way no matter where tracing starts or if the shape is flipped.
3. During training, the AI compares its predicted distance table against the real one. It automatically finds the best-matching starting point and enforces basic geometry rules—like making sure distances are never negative and that a point's distance to itself is zero.
4. To convert the AI's compact code back into a visual drawing, ShapeEmbed takes the predicted distance table and uses Multidimensional Scaling (MDS) to plot the points back into 2D space.

5.7 Feature Selection and Whole-Embryo Vectorization

Cell-level morphological measurements from all segmented embryos were compiled into a unified dataset for feature filtering and model training. To eliminate multi-collinearity and reduce feature redundancy, I constructed a pairwise Pearson correlation matrix across all 16 extracted parameters. A correlation threshold of $|r| > 0.75$ was established; any feature exhibiting a correlation greater than 0.75 with a preceding feature was considered redundant and removed. Crucially, the raw correlation between **Angle** and **FeretAngle** appeared artificially low due to angular periodicity near boundary values ($0^\circ \equiv 180^\circ$, Fig.). Rearranging the angular metrics to account for circular continuity revealed strong correlation as expected (Fig.). Following correlation filtering, four mutually independent core features were retained for downstream analysis: cell area (**Area**), orientation angle (**Angle**), circularity (**Circ.**), and solidity (**Solidity**).

To enable machine learning classifiers to evaluate tissue-scale phenotypes rather than individual cells, single-cell measurements were aggregated into whole-embryo profile vectors. For each embryo, a 16-bin fixed-width histogram was generated for each of the four core features. The bin heights across all four distributions were concatenated sequentially, yielding a single 64-dimensional summary vector (4 features $\times$ 16 bins) per embryo to serve as the input representation for the Random Forest and Support Vector Machine models.





(a) Gel electrophoresis result of del15 PCR. (b) Binary masks of segmented cells. (c) Rotated image with a horizontal midline. (d) Cropped image used for analysis. (e) Fixed image of a stage 7 yw embryo for neighbour counting. (f) Boundary graph for the same embryo. (g) Centroid graph for the same embryo. (h) del15 deletion. (i) Colour profile of a line crossing the boundary once.

5.8 Leave-One-Out Cross-Validation

Due to the constrained sample size ($N = 13$ live-imaged embryos), conventional train–test splits would substantially reduce the training set size and introduce sampling bias. To rigorously evaluate the generalization performance of both the Random Forest and Support Vector Machine models without losing valuable training data, I implemented a Leave-One-Out Cross-Validation (LOOCV) scheme.

In each iteration of LOOCV, a single embryo vector was held out as an unobserved test sample, while the remaining $N - 1$ embryo vectors were used to train the model. This procedure was executed iteratively N times, ensuring that every embryo in the cohort served as the test subject exactly once. Model performance was evaluated based on the aggregate classification accuracy across all N validation folds, yielding a robust estimate of classifier reliability on small datasets.

5.9 Neighbour Counting Algorithm

To evaluate local cell-cell connectivity and tissue topology, I developed an automated neighbour-counting algorithm based on dual-mask ray-tracing analysis. Segmented cell outlines were rendered as a binary mask and skeletonized to a uniform 1-pixel-thick boundary network. Simultaneously, a secondary coordinate mask was generated containing discrete centroid markers of every cell.

For each target cell, its skeletonized boundary was dilated outward by 30 pixels to establish a localized candidate search region. Candidate neighbouring centroids falling within this dilated region were identified using local maxima detection on the centroid mask. To verify true topological adjacency, straight line segments were projected from the target cell centroid to each candidate centroid across the skeletonized boundary mask. The pixel intensity profile along each projected segment was evaluated to tally boundary line intersections. Candidates were validated as direct physical neighbours if the connecting segment crossed ≤ 1 boundary line—a criterion that permits 0 crossings to accommodate minor local segmentation breaks—and their pairwise Euclidean distances were recorded.

Acknowledgements

I am very thankful to my supervisors, Jenny and Liz, for their continuous guidance, expertise, and support throughout this project. I am deeply grateful to Jenny for her hands-on training and patient mentorship across all the experimental work. Equally, I want to thank Liz for her dedicated support through the dry lab and computational analyses.

Finally, I would like to express my sincere gratitude to Professor Sanson and the entire Sanson Lab for giving me this invaluable opportunity to undertake this research project. Thank you to all the members of the lab for fostering such a welcoming and collaborative environment.

References

- [1] Finegan, T.M., Hervieux, N., Nestor-Bergmann, A., Fletcher, A.G., Blanchard, G.B. and Sanson, B. (2019). The tricellular vertex-specific adhesion molecule sidekick facilitates polarised cell intercalation during drosophila axis extension. *PLOS Biology*, 17(12), p.e3000522. doi:10.1371/journal.pbio.3000522.
- [2] Foix Romero, A., Russell, C., Krull, A. and Uhlmann, V. (2025). ShapeEmbed: A self-supervised learning framework for 2D contour quantification. *arXiv*. Available at: <https://arxiv.org/abs/2507.01009v1> [Accessed 1 Sept. 2026].
- [3] Lye, C.M., Naylor, H.W. and Sanson, B. (2014). Subcellular localisations of the CPTI collection of YFP-tagged proteins in *Drosophila* embryos. *Development* (Cambridge, England), [online] 141(20), pp.4006–17. doi:10.1242/dev.111310.
- [4] Yamagata, M. (2020). Structure and functions of sidekicks. *Frontiers in Molecular Neuroscience*, 13. doi:10.3389/fnmol.2020.00139.
- [5] Park, J.-A., Kim, J.H., Bi, D., Mitchel, J.A., Qazvini, N.T., Tantisira, K., Park, C.Y., McGill, M., Kim, S.-H., Gweon, B., Notbohm, J., Steward, R., Burger, S., Randell, S.H., Kho, A.T., Tambe, D.T., Hardin, C., Shore, S.A., Israel, E., Weitz, D.A., Tschumperlin, D.J., Henske, E.P., Weiss, S.T., Manning, M.L., Butler, J.P., Drazen, J.M. and Fredberg, J.J. (2015). Unjamming and cell shape in the asthmatic airway epithelium. *Nature materials*, [online] 14(10), pp.1040–8. doi:10.1038/nmat4357.